

Maneuvering Formations in Directional Fields: Path-Integrated Exposure and Delayed Threats in Nonholonomic Adversarial Games

Research Layer of the *Iron Bottom Sound* project*

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Abstract

We study adversarial trajectory optimization between two moving *formations* that interact through a directional field under nonholonomic motion constraints. The decision variable is not a position but a trajectory, and the objective is the path-integrated net exposure built from a calibrated anisotropic interaction kernel; we characterize the resulting closed-loop dynamic game and test its mechanisms numerically. Three results are proved and measured. First, the path-integrated objective is load-bearing: snapshot and peak positional surrogates reorder committed plans and can reverse a whole-trajectory ranking in a constructed example. Second, a symmetric closed-loop degeneracy: under explicit player-exchange symmetry the game value collapses exactly to the instantaneous payoff, while capability monotonicity holds for nested maximum-speed sets. Third, formation representation matters as a robustness layer: the numerical naval instantiation uses a leader–follower line-ahead formation in which trailing vessels follow the leader’s track at fixed arc-length offsets, and the comparative statics tested here are unchanged when the rigid baseline is substituted. We also report a matched-horizon replanning experiment that fixes the evaluation horizon and varies only the replanning cadence: the earlier, horizon-confounded indication that longer commitment favors the range-advantaged side does *not* survive the correction, so we remove commitment as a claim and report the corrected experiment as a negative result with its control diagnostic. Delayed threats act through response-space contraction, with coverage and local intensity interacting rather than either acting alone. Finally, a matched external test in an auditable rules engine shows that transfer of the reduced model’s point values is partial, which locates the boundary of the reduced model and makes the engine an external case study rather than a validation claim.

1 Introduction

How should an adversarial formation maneuver when the interaction strength is directional, the motion is nonholonomic, and some controls have delayed effects? The classical naval answer — cross the enemy’s line to concentrate broadside fire while presenting only

*All experiments are reproducible from repository scripts against a frozen deterministic rules engine (commit `fc77a65`). The WWII surface-combat wargame serves only as an external case study; this paper’s claims are about the general model class.

bow or stern arcs — is a doctrine about *positions*. This paper argues that positions are the wrong primitive. What matters is the *cumulative interaction along trajectories*: the path-integrated net exposure an opponent suffers versus the exposure one must sustain in return. From this vantage, crossing the T is merely one geometric state that may or may not raise the integral — never an objective in itself.

We develop this view on a complete, auditable wargame engine (deterministic conditional on sealed orders and random seed) (*Iron Bottom Sound*, IBS): a hexagonal WWII surface-combat simulation whose adjudication is a pure function of state, sealed orders, and seed, with every die roll audited against its rule reference. The engine provides the calibration target for our interaction field and the case-study environment for our policies; the mathematical claims are about the model class, not the wargame. The resulting pipeline is:

1. calibrate a smooth anisotropic interaction kernel against the engine’s exact adjudication (retained fidelity $\rho = 0.991$);
2. embed the kernel in a committed-maneuver dynamic game over path-integrated exposure;
3. characterize the structural properties (the degeneracy boundary and speed-capability monotonicity) and test a commitment-type comparative static that the fixed-horizon correction subsequently removes;
4. translate model policies back into legal engine orders and validate against doctrine baselines with paired seeds.

Each step is gated by pre-specified acceptance criteria and every number in this paper traces to a script in the repository.

1.1 Contributions

1. **A path-integrated anisotropic formation-game formulation** for moving formations ((1)–(3)): we show that snapshot positional quality can reorder committed plans and can reverse a whole-trajectory ranking, so the integral — not the best position — is the object of optimisation. The interaction kernel is calibrated against an auditable rules engine ($\rho = 0.991$, nMAE 5.7%), and an isotropic ablation shows that removing directionality reduces held-out rank correlation from 0.991 to 0.952.
2. **Structural closed-loop results under explicit symmetry assumptions**: a symmetric closed-loop degeneracy with a complete proof (skew-symmetric successor matrices collapse the value onto the instantaneous payoff; the collapse is an interaction-symmetry phenomenon, untouched by kinematic asymmetry), and maximum-speed capability monotonicity by nesting.
3. **A delayed spatiotemporal threat formulation** in which response-space contraction depends jointly on threat coverage and local intensity, with the near-optimal response set as the mediating object.

The leader–follower formation used for the numerical instantiation is a fidelity layer rather than an algorithmic contribution, and the rules-engine study is an external case study, not a validation claim. We additionally report, as a negative result, that a commitment–range comparative static which appears in a horizon-confounded open-loop experiment does not survive a matched-horizon replanning experiment (Section 6).

2 Related Work

2.1 Dynamic and differential games

Zero-sum differential games originate with Isaacs’ method of characteristics [3]; the theory of feedback (closed-loop) versus open-loop information structures and their associated Nash values is developed in [4]. Classical pursuit–evasion and combat models optimize terminal capture, separation, or energy-style objectives; Lanchester-type attrition and its optimal-control treatment [14] and PDE combat models [15] aggregate force levels rather than adjudicate anisotropic single-weapon interactions. Our point of departure is the objective: a *path-integrated anisotropic interaction* — the cumulative net exposure along whole trajectories — for which instantaneous geometry is only an ingredient, not the prize.

2.2 Reachability and viability

Reachable-set and viability-kernel methods characterise states a system can maintain under constraints [5], with Hamilton–Jacobi reachability providing adversarial variants for continuous dynamic games [6]. In that literature the target set is given ex ante. Here the “favorable region” is itself defined by the directional interaction value ($L \geq \alpha$), and reachability questions are coupled to a path-integrated payoff: we ask not only whether a state is reachable or invariant, but what sustaining it is worth along the trajectory.

2.3 Commitment and information structure

The strategic value of binding oneself — committing to a plan the opponent can observe — is classical [7]; in dynamic games it separates open-loop from feedback equilibria [4], and receding-horizon control makes the replanning cadence a design variable [8]. Algorithmically, our open-loop solver is a double-oracle scheme in the spirit of [9]. What this paper adds to that literature is the treatment of *formation-level* trajectories in an adversary’s directional field with a path-integrated payoff, together with an explicit negative finding about a commitment-type comparative static: an effect that appears when plan length and evaluation horizon vary together does not survive once the horizon is held fixed (Section 6).

2.4 Leader–follower formation control

Path-following formation control and vehicular platooning study exactly the rule used here as the naval instantiation: a designated leader generates a track and followers regulate along-path offsets and adopt the local tangent [1, 2]. That literature asks how to track a given path with bounded spacing error; we use the same representation but ask a game-theoretic question — which path to choose when an adversary’s directional field and delayed weapons make the payoff path-integrated. The distinction matters for interpretation: fixed-offset and path-following representations are *different formation models*, not two discretisations of one model, which is why the robustness section reports both rather than assuming equivalence.

2.5 Interdiction and spatiotemporal hazards

Time-dependent shortest paths [10] and network interdiction [11] study adversaries who add costs to arcs or destroy links. Robust path planning around dynamic obstacles treats hazards as spatial fields to avoid. The delayed threat studied here differs mechanically: a launched weapon is a *stateful spacetime field* that does not merely add path cost but reshapes the opponent’s set of near-optimal future responses — a response-set contraction measured against the near-optimal landscape, not against feasibility.

3 General Formation Game

A player controls a formation, not a vessel. Throughout, each player $i \in \{B, R\}$ commands a moving *formation*: a designated leader state together with a formation map that places the remaining vessels. Write $\mathcal{H}_i(t)$ for the leader history (position, heading, speed and the path traversed so far) and let vessel k of formation i occupy

$$X_{ik}(t) = \mathcal{G}_{ik}(\mathcal{H}_i(t)), \quad (1)$$

for a formation map $\mathcal{G}_{ik}$. The leader obeys nonholonomic kinematics, $\dot{x}_i = v_i \cos \psi_i$, $\dot{y}_i = v_i \sin \psi_i$, $\dot{\psi}_i = \omega_i$, with $v \leq \bar{v}_i$ and $|\omega_i| \leq \bar{\omega}$ (a minimum turn radius $R_{\min}(v) = v/\bar{\omega}$ couples speed and turning). Lateral displacement therefore requires first rotating the velocity vector — the defining nonholonomic cost.

Equation (1) deliberately leaves the formation map free. Two instantiations are used in this paper:

$$\underbrace{\mathcal{G}_{ik}^R = c_i + R(\psi_i) p_{ik}}_{\text{rigid baseline (fixed offsets)}}, \quad \underbrace{\mathcal{G}_{ik}^{\text{LF}} = \gamma_i(s_i - \ell_k)}_{\text{leader-follower (path following)}}, \quad (2)$$

where γ_i is the leader’s arc-length parameterised track, s_i the leader’s arc length and ℓ_k the vessels’ arc-length offsets. The rigid map is analytically transparent and is treated as a baseline; the path-following map is the main numerical instantiation (Section 9), because a line-ahead column of real ships follows the leader’s track through a turn instead of rotating as one body. The structural results below are stated for the general map: they require only the stated symmetry assumptions on the induced game, not a particular $\mathcal{G}$. No experiment in this paper supports a claim about isolated single vessels.

Anisotropic formation field. Each vessel projects an interaction kernel that depends on range, relative bearing, the two vessels’ headings, and capability state; because vessels of the same formation may carry different headings under $\mathcal{G}^{\text{LF}}$, the kernel is indexed by the shooter’s own heading ψ_{ik} . The formation field aggregates over its vessels,

$$F_i(x, t) = \sum_{k \in i} K_{ik}(x, t), \quad (3)$$

and formation-vs-formation interaction sums over vessel pairs. In the naval instantiation K is the calibrated kernel of Section 9.1 and each formation is a line-ahead column.

Proposition 1 (Formation aggregation). *For rigid formations, formation-to-formation exposure is exactly additive over vessel pairs:*

$$E_{i \rightarrow j} = \int_0^T F_i(x_j(t), t) dt = \sum_{k \in i} \sum_{m \in j} \int_0^T K(x_{jm}(t); x_{ik}(t), t) dt.$$

Proof. By (3) the formation field is a sum of per-vessel kernels, and the integral is linear. Hence the exposure of vessel m of formation j equals the sum, over the vessels k of formation i , of the pairwise terms; summing over $m \in j$ gives the stated identity. $\square$

This elementary identity pins down what “formation payoff” means operationally: the double sum over vessel pairs that all solvers in this paper evaluate.

Path-integrated objective. Against trajectory $x_j(t)$, the cumulative exposure is $E_{B \rightarrow j} = \int_0^T F_B(x_j(t), t) dt$, and the net path value is

$$J_{\text{gun}} = E_{B \rightarrow R} - \lambda E_{R \rightarrow B}, \quad \lambda = 1. \quad (4)$$

The object of optimisation is not a position but a trajectory: the integral is the primitive. Discretely, $J = \sum_t \Delta t [P_{B \rightarrow R}(t) - \lambda P_{R \rightarrow B}(t)]$ with formation-aggregate P from (3).

Delayed threat state. Torpedoes are not damage terms but a delayed stateful field: the state extends to $z_t = (s_t, q_t)$, where q_t records every launched weapon’s position, heading, speed, remaining range, and threat probability. The spatiotemporal hazard $H(x, t; q_t)$ is generated by past launches and constrains future responses (Section 8).

Information and commitment. In the closed loop both sides re-optimize every step against current state. In the committed (open loop) game both sides seal formation heading plans of horizon H before execution — mirroring the engine’s sealed-order discipline.

4 Structural Properties of the Closed-Loop Game

4.1 Closed-loop degeneracy and its boundary

Proposition 2 (Symmetric closed-loop degeneracy). *Consider a two-player zero-sum dynamic game with simultaneous moves, a common finite action set U with mixed play on the simplex $\Delta(U)$, a stage payoff that depends on the state only, $g(s, u_B, u_R) = \ell(s)$, and a deterministic successor map F . Write $\mathcal{S}$ for the player-exchange map on states (the relabelling that swaps which formation is B and which is R). Assume:*

(A1) odd stage payoff under exchange: $\ell(\mathcal{S}s) = -\ell(s)$;

(A2) equivariant dynamics: $\mathcal{S}F(s, u_B, u_R) = F(\mathcal{S}s, u_R, u_B)$.

Then at every state s the one-step successor-payoff matrix $A(s) = [\ell(F(s, u, v))]_{u, v \in U}$ is skew-symmetric, and:

- (i) finite horizon H with any $\gamma \leq 1$: the value function obtained by backward induction from any terminal value satisfying $V_{H+1}(\mathcal{S}s) = -V_{H+1}(s)$ equals the instantaneous payoff, $V_h \equiv L \equiv \ell$;
- (ii) infinite horizon with $0 < \gamma < 1$: the Bellman operator T is a γ -contraction on bounded functions, L is its fixed point, and $V^* \equiv L$ by the contraction mapping theorem.

Proof. Skew-symmetry. Take $s, u, v \in U$. By (A2), $F(s, u, v)$ and $\mathcal{S}F(s, v, u)$ denote the same state, so $A_{uv}(s) = \ell(F(s, u, v)) = \ell(\mathcal{S}F(s, v, u)) = -A_{vu}(s)$ by (A1). Hence $A(s)^\top = -A(s)$.

Stage value. For skew A and any $x \in \Delta(U)$, $x^\top A x = \sum_{u, v} x_u x_v A_{uv} = 0$: diagonal terms vanish ($A_{uu} = 0$) and off-diagonal terms cancel in pairs. Thus $\min_y x^\top A y \leq$

A line-ahead formation bends along the leader's track;
the payoff integrates exposure over whole trajectories

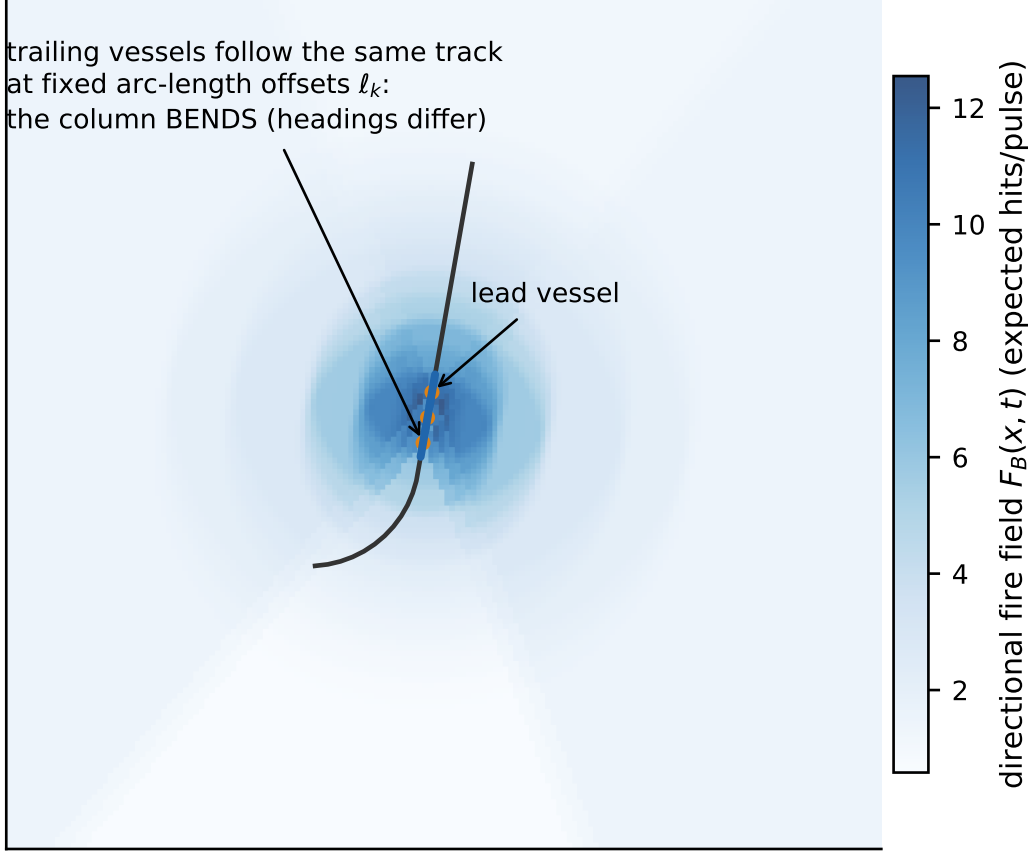


Figure 1: A line-ahead formation manoeuvring inside a directional fire field: the leader generates a track and the trailing vessels follow it at fixed arc-length offsets, so the column *bends* through the turn and the vessels carry different headings. The payoff integrates exposure along whole trajectories; a delayed torpedo hazard would add a stateful spacetime field on top of the instantaneous field shown here.

0 for all x , so $\max_x \min_y x^\top A y \leq 0$; and $y^\top A y = 0$ gives $\max_x x^\top A y \geq 0$ for all y , so $\min_y \max_x x^\top A y \geq 0$. By the minimax theorem both equal the same value, hence $\text{val } A(s) = 0$ for every s .

(i) *Backward induction*, $\gamma \leq 1$. Suppose $V_{h+1}(\mathcal{S}s) = -V_{h+1}(s)$. The continuation matrix at s is $A^{(h+1)}(s) = [\gamma V_{h+1}(F(s, u, v))]_{u,v}$. By (A2) and the induction hypothesis it is again skew-symmetric, so its minimax value is 0 and $V_h(s) = \ell(s) + \text{val } A^{(h+1)}(s) = \ell(s) = L(s)$, restoring the hypothesis one step earlier. Induction down to $h = 0$ gives $V_h \equiv L$ for every horizon.

(ii) *Infinite horizon*, $0 < \gamma < 1$. For bounded V, V' , $|\max_x \min_y x^\top A[V]y - \max_x \min_y x^\top A[V']y| \leq \max_{u,v} |V(F(s, u, v)) - V'(F(s, u, v))| \leq \|V - V'\|_\infty$ because $\max_x \min_y$ is 1-Lipschitz in the matrix entries under the sup-norm; hence T is a γ -contraction. Since $(TL)(s) = \ell(s) + \gamma \cdot 0 = L(s)$, L is a fixed point, and uniqueness of the fixed point gives $V^* = L$. $\square$ $\square$

Remark 1 (Scope of the skew-symmetry condition). (A1) is not implied by identical hardware alone; it is a statement about the exchange symmetry of the closed loop in the

relative state. It holds exactly at player-exchange-symmetric configurations when the stage payoff is odd under the exchange and the successor map is equivariant, and in the calibrated symmetric model it holds at machine precision across the full state grid ($\max |V - L| = 0$ on two grid sizes, two value-iteration sweeps; exchange-symmetry error 0). Our solver uses the pure-strategy Bellman recursion; it shares the fixed point because the pure max-min of $A(s)$ is also 0 on the entire symmetric grid. The boundary of the proposition is measured in Table 1.

The interesting science is the *boundary* of the proposition. We break each assumption and re-solve (Table 1):

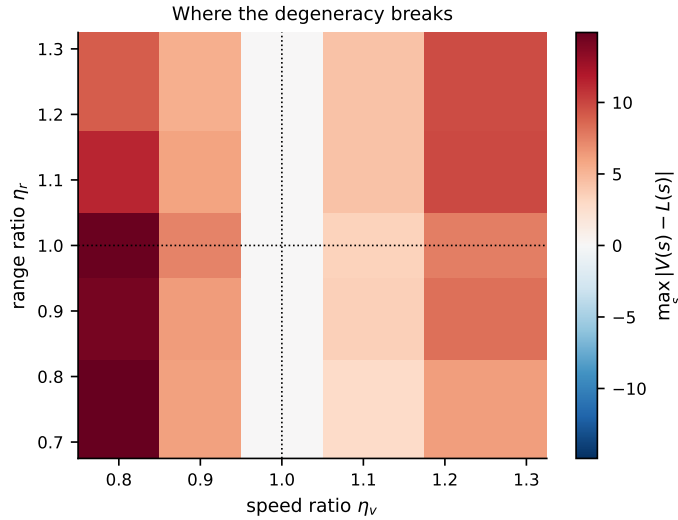


Figure 2: Where the degeneracy breaks. Colour is $\max_s |V(s) - L(s)|$, the closed-loop excess value over the instantaneous payoff, on the (η_v, η_r) grid: the *horizontal* axis is the speed ratio (speed asymmetry) and the *vertical* axis is the range ratio (range / interaction-envelope asymmetry). In the tested perturbations the range asymmetry breaks the collapse whereas the tested speed asymmetry leaves it essentially intact.

Table 1: Assumption-breaking matrix: $\max |V - L|$ after re-solving the closed loop with one assumption violated at a time.

Violated assumption	$\max V - L $	degeneracy
(none — symmetric base)	0.0	held
speed asymmetry ($\eta_v = 1.25 / 0.80$)	0.0 / 0.0	held
action asymmetry (5 turn levels vs 3)	0.0	held
symmetry-preserving terminal payoff	0.0	held
range asymmetry ($\eta_r = 1.25$ vs 0.80)	13.99	broken
firepower asymmetry ($\eta_g = 1.25$)	20.09	broken
all combined	9.75	broken

The pattern is sharp in the tested perturbations: *kinematic* asymmetries (speed, action sets) and the tested *symmetry-preserving* terminal-payoff perturbation leave the degeneracy exactly intact, while *interaction-strength* asymmetries (range, firepower) destroy it. The degeneracy is an interaction-symmetry phenomenon: kinematic asymmetries alone

did not unlock positional value in the tested conditions; only interaction-strength asymmetries did. This localized prediction — and its contrast with the committed game below — is the paper’s central structural finding.

4.2 Finite-horizon DP refinement

The plan-library approach of Section 6 suffers from a first-order limitation: the committed-maneuver values depend on the discretisation of the plan library. A nested-library test (11 → 21 → 41 plans) shows convergence in only 25% of cells (gate $\geq 80\%$) and maximin plan stability in 18% (gate $\geq 85\%$). We therefore replace the plan library with a *finite-horizon backward induction* solver on the reduced-state grid:

$$V_T(s) = 0, \quad V_t(s) = \max_{u_B} \min_{u_R} [L(s) \Delta t + V_{t+1}(F(s, u_B, u_R))], \quad (5)$$

with three grid refinements (coarse 16×12 , medium 30×24 , fine 48×36). Across seven representative test cells (symmetric head-on/parallel, range advantage/disadvantage, crossing, speed advantage), **all cells converge** ($\delta_V < 5\%$; most $< 2\%$). The solver confirms: (i) symmetric configurations yield $V \approx 0$ (the degeneracy prediction of Proposition 2); (ii) range asymmetry produces strongly signed values (± 8); (iii) speed asymmetry alone does not break the degeneracy (consistent with the interaction-symmetry boundary). The plan-library approach is retained only for the *committed policy* translation layer (Section 11). An earlier exploratory engine comparison motivated the later matched transfer study.

Plan-library convergence is a first-order concern for committed-maneuver studies, and our nested-library test returns an honest failure: enlarging the library from 11 to 21 to 41 plans changes the maximin value by more than the 5% convergence gate in 75% of cells (gate: $\geq 80\%$ converged), and the maximin plan identity is stable in only 18%. Per the plan’s decision tree, we therefore (i) do *not* describe the phase diagrams as converged equilibrium regimes but as *library-approximate* regime maps; (ii) report all values as the interval $[V_{21}, V_{41}]$; and (iii) verify that the robust conclusions (the sign structure from range asymmetry, the swap symmetry, the degeneracy boundary of Table 1) are library-independent — they are propositions about the model, not outputs of the library. The library-approximate maps over (η_v, η_r) and their regime labels are given in the repository; the regime *ordering* (which asymmetry dominates) is stable even where exact values are not. The lesson for the method class: committed-maneuver equilibrium values are *library-sensitive* in a way that the structural propositions are not, and any committed-game paper should publish its library-sensitivity audit alongside its values.

4.3 Speed-capability monotonicity

Proposition 3 (Capability monotonicity). *Let $V(\bar{v})$ denote the game value when player B’s admissible closed-loop policy set is $\mathcal{P}(\bar{v})$ — the policies whose action at every information state lies in the capability set $[0, \bar{v}]$ (with a common turn-rate bound). If $\mathcal{P}(\bar{v}_1) \subseteq \mathcal{P}(\bar{v}_2)$, then $V(\bar{v}_2) \geq V(\bar{v}_1)$.*

Proof. $V(\bar{v}) = \max_{\pi \in \mathcal{P}(\bar{v})} \min_{\pi_R} J(\pi, \pi_R)$. Taking the maximum over a superset cannot decrease it. $\square$ $\square$

Capability vs. operating speed. The premise requires *nested capability* semantics — a vessel’s cap $\bar{v} \in \{2, 4, 6\}$ admits speed tiers $\{v \leq \bar{v}\}$, giving $U(2) \subset U(4) \subset U(6)$.

A *fixed operating speed* $v = \bar{v}$ changes the dynamics non-nestedly and the proposition makes no claim about it; conflating the two produced what earlier drafts called a “speed paradox,” a term we retire. Measured over two geometries $\times$ three range ratios: the open-loop maximin value is non-decreasing in $\bar{v}$ in every cell (e.g., parallel geometry at $\eta_r = 1$: $V = -0.571 \rightarrow -0.135 \rightarrow +0.000$; at $\eta_r = 1.33$: $+2.66 \rightarrow +2.69 \rightarrow +2.77$); the largest single-step “drop” across all cells is -2.2×10^{-15} (LP noise). **Speed-capability monotonicity holds.** The contrast with the fixed-operating-speed sweep — where the value *decreases* with speed in most geometries — confirms that the earlier “speed paradox” is purely an artifact of the operating-speed semantics: given the *capability* to choose, more speed never hurts; forced to *operate* at higher speed without a positional objective, it hurts (supplement).

5 Path-integrated Formation Maneuver

5.1 The path-integrated objective is load-bearing

For identical candidate-trajectory sets we compare three objectives: the terminal snapshot $L(s_T)$, the peak $\max_t L(t)$, and the integral J . Across nine parameter cells ($\eta_v \in \{0.8, 1.0, 1.25\} \times$ three geometries; 1,089 rollouts):

- Kendall τ between integral and snapshot rankings: mean 0.476 for $L(s_T)$ (95% CI $[0.404, 0.548]$) and 0.365 for $\max_t L$ — far from the $\tau \geq 0.95$ equivalence band; **no cell is rank-equivalent**.
- The peak objective’s worst-case vector degenerates completely in 9/9 cells (a best-responding defender holds $L(t) \leq 0$ at every t): under maximin, the peak objective has *no discriminating power*.
- Constructed reversal (real rollouts): trajectory A (turn $+180^\circ$ into a crossing shot) reaches a peak $L = 2.88$ at $t = 2$ but integrates to $J = 1.32$; trajectory B (turn $+30^\circ$, sustained oblique) peaks at 0.29 but integrates to $J = 5.02$. The peak criterion and the integral criterion prefer opposite trajectories.

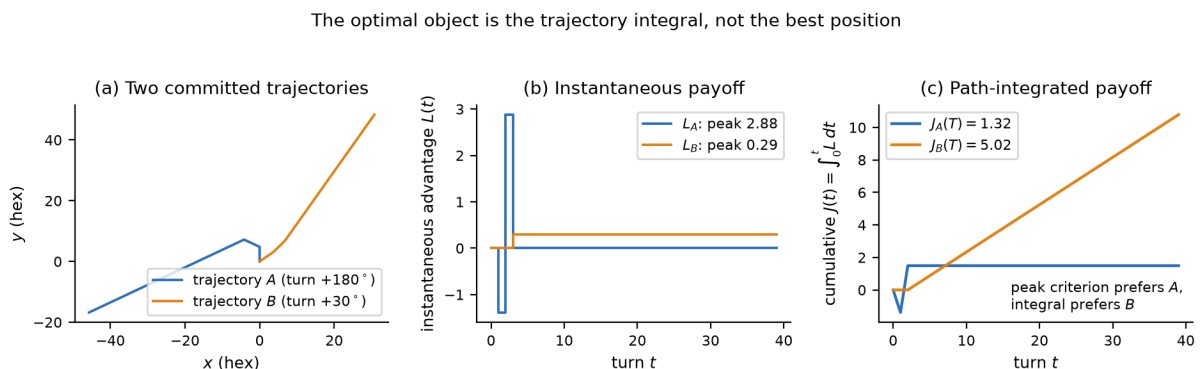


Figure 3: Snapshot versus integrated objectives on two real committed trajectories. (a) The plans and their paths. (b) Instantaneous advantage: trajectory A reaches a much higher peak. (c) Cumulative exposure: the integral prefers the sustained moderate posture.

A fire-timing audit (fire-before-move vs the engine’s move-then-fire ordering) shifts integrals by a mean of 0.08 with *zero* sign changes across 1,089 plan pairs: the ordering offset is second-order. Fleet-level exposure maps show the column’s mid-flank sustained +18.8% higher directional density than the van flank — a structural field property that only appears once the field, not the position, is the object.

5.2 Lateral reposition cost

A finer $d_\perp \in \{0, \dots, 10\}$ sweep reveals the cost’s structure: $C_{\text{reposition}}$ starts at +2.9 for $d_\perp \approx 0$ (where the fire is most concentrated), decays to +0.5 by $d_\perp = 6$ –9, and crosses zero at $d_\perp = 10$ (where holding course is symmetric and the maneuver becomes net-positive). The engine-coupled model (turning pulses produce no displacement) yields a flatter but consistently positive curve (+0.55 to +0.69 within range). The nonholonomic tax is a real, range-dependent levy on turning, and it is precisely the kind of cost that the corrected replanning experiment of Section 6 probes; the representative curve is given in the supplement.

6 Commitment, Replanning Cadence, and a Corrected Experiment

6.1 The earlier open-loop experiment and its confound

An earlier experiment in this project compared open-loop games that *sealed* plans of length H and accumulated the payoff over the same H stages, reporting $P_H = V_H - V_1$. That comparison changes two things at once — how long a plan is locked, and how many stages are evaluated — so it cannot be read as the value of commitment. We therefore retain its numbers only as an *open-loop engagement-horizon effect*: with a longer engagement horizon the cumulative value shifts systematically, and in the library-based and continuous-oracle variants of that experiment the sign of the shift tracked the range advantage. The corresponding certified signs of that metric are reported in the supplement; they are **not** used as a commitment claim anywhere in this paper.

6.2 A matched-horizon replanning experiment

Definition. Fix the evaluation horizon $T = 6$ for every policy class and vary only the *replanning cadence* $h \in \{1, 2, 3, 6\}$: with cadence h the state is observed, the remaining-horizon open-loop game is solved, and the resulting plan is locked for h steps before the state is re-observed. $h = 1$ is maximum flexibility and $h = 6$ is full commitment. The stage reward is the trapezoid interval reward $r_t = \frac{1}{2}[L(s_t) + L(s_{t+1})]$, so an action always affects the reward of the step in which it is executed (this retires an artefact of the earlier formulation, in which a one-step policy’s action could be payoff-neutral). The commitment metric is

$$C_h = V_T^{(h)} - V_T^{(1)}, \quad C_1 = 0, \quad (6)$$

so $C_1 = 0$ by construction and the comparison is between cadences, not horizons.

Implementation. A single planner is used for every h : same leader–follower dynamics, same field, same reward, same horizon, same action grid (Grid-5), same equilibrium solver, same discount ($\gamma = 1$); the only difference is the replanning interval. At each replanning

epoch the remaining-horizon open-loop game is solved with the restricted-matrix plus exhaustive-best-response scheme of Section 6, and *both sides sample from the complete equilibrium mixture* — no argmax action, no averaged action, no support truncation. Sampled plans are evaluated with paired seeds (common random numbers), and C_h is reported with a paired bootstrap confidence interval. Because the planner is deterministic given the state and randomness enters only through mixture sampling, these values are Monte Carlo expectations of this receding implementation, not an exact extensive-form feedback equilibrium.

Pre-specified criterion. Before the run we fixed the criterion: the effect would be retained as a claim only if the C_6 confidence interval excluded zero in at least 15 of the 18 asymmetric cells with $\text{sign}(C_6) = \text{sign}(\eta_r - 1)$ in those cells, with both range directions supported in every geometry and no significant reversal in the Grid-7 anchors; a partial outcome in 10–14 cells would have been reported as conditional; and more than two strongly significant reversals would force removal of the claim.

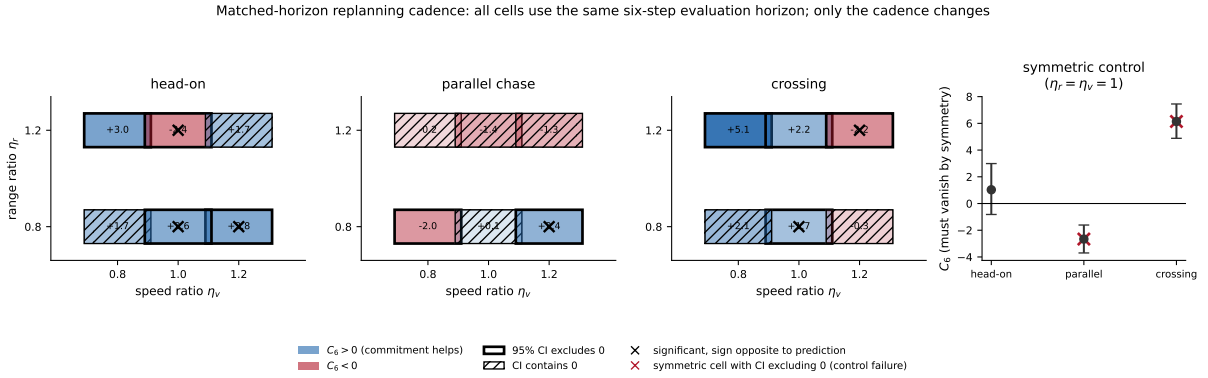


Figure 4: Matched-horizon replanning-cadence experiment. *All cells use the same six-step evaluation horizon; only the replanning cadence changes* ($h = 1$ maximum flexibility versus $h = 6$ full commitment); colour is the paired mean C_6 and a solid border marks a paired bootstrap interval excluding zero. A cross marks a significant cell whose sign contradicts $\text{sign}(\eta_r - 1)$. The rightmost panel is the symmetric control ($\eta_r = \eta_v = 1$), where C_6 must vanish by symmetry: it does so in head-on but not in the other two configurations, which bounds the residual bias of this receding implementation.

Result: the effect does not survive. Across the 18 asymmetric cells with 200 paired seeds each, 10 cells have a C_6 interval excluding zero, and *only four of those ten have the sign predicted by the range advantage*: six are significant reversals. The geometry support is fragmentary rather than bidirectional (head-on 1 positive and no negative; parallel none positive and one negative; crossing two positive and none negative). Under the pre-specified criterion this forces removal of the claim: with the evaluation horizon held fixed and only the replanning cadence varied, the commitment–range comparative static does not reproduce.

Control diagnostic (why we treat this as a null rather than a reversal). In a symmetric configuration ($\eta_r = \eta_v = 1$) the metric must vanish by symmetry, and it does so in head-on ($C_6 = +1.0$, $[-0.8, +3.0]$, interval containing zero) but *not* in crossing ($+6.1$, $[+4.9, +7.5]$) or parallel (-2.7 , $[-3.7, -1.6]$). In crossing the payoff is provably antisymmetric in this configuration ($\max |J(p, q) + J(q, p)| = 0$ and $J(p, p) = 0$ to machine precision), so the residual cannot be a model asymmetry: it comes from the receding solver playing an equilibrium of a *restricted* plan set, whose value can be exploited by

plans outside that set. The bias is of the same order (1–6) as the effects measured in the asymmetric cells. We therefore report the matched-horizon experiment as an inconclusive-to-negative result: the corrected metric neither confirms the earlier pattern nor supports a reversed one, and commitment is removed from the paper’s claims.

What is retained. The honest conclusion is methodological and is stated as such: (i) the earlier commitment numbers are horizon effects and must be renamed; (ii) under a matched horizon the effect does not reproduce at this implementation fidelity; and (iii) certifying a receding policy requires a full-space equilibrium, which the restricted-plan-set solver cannot deliver — an object for future work rather than a claim here.

7 Reachability and Viability

7.1 No guaranteed favorable state in open water

With the favorable set $G_\alpha = \{s : L(s) \geq \alpha\}$ defined relative to each configuration’s co-operative peak ($\alpha = \tau = 0.5$), we ask whether the committed maximin plan guarantees reachability (worst-case peak $\geq \alpha$) or viability (worst-case holding share $\geq \tau$) against *all* eleven defender plans, over 144 cells (three geometries $\times$ sixteen speed ratios $\times$ three turn scalings). **No cell admits a guarantee.** The mechanism is the defender’s standing *decline-battle option*: in open water a best-responding opponent can disengage at will, so the worst-case advantage-retention ratio never reaches α and the per-plan retention landscape collapses to a median of 0–3% of steps. Per the plan’s decision tree we therefore pivot from threshold-hunting to the viability landscape (reported as per-plan quantiles in the repository). This negative result sharpens the commitment finding: positional guarantees are purchasable only through range/firepower asymmetry (which shifts L itself) or through the opponent’s own commitments.

8 Delayed Threats and Response-Space Contraction

8.1 The delayed threat state

Torpedoes extend the state to $z = (s, q)$. Each launched weapon carries its analytic (hex, impulse) timeline — verified cell-by-cell against a real engine adjudication — so the hazard $H(x, t; q)$ genuinely depends on time: the same hex is threatened at one impulse and safe at another quantified in Figure 5(a,b).

8.2 Response-set contraction

Let $\mathcal{B}_\epsilon = \{\tau : J(\tau) \geq J^* - \epsilon\}$ be the ϵ -near-optimal response set over the target’s 340–600 legal plans, and $C_{\text{resp}} = 1 - |\mathcal{B}_\epsilon^H|/|\mathcal{B}_\epsilon|$ its contraction under the threat. The response-set experiment measures C_{resp} over the defender’s complete plan space (340–600 plans) as a function of threat thickness k . At the 5% response-set threshold, the mean contraction increases with salvo count in the unmatched-lethality sweep: $C_{\text{resp}} = 0.20 \rightarrow 0.31$ from $k = 1$ to $k = 6$, with individual cells reaching full contraction ($C_{\text{resp}} = 1.0$) when the threat saturates the top tier. In the low-hit regime ($k=1$), the contraction is exactly zero in 7 of 8 cells — reproducing the structural finding of the low-hit regime — while thicker salvos progressively remove tied top-tier routes. The single exception (a close-range pair with a two-plan top tier) has the entire top tier covered by one track, yielding $C_{\text{resp}} = 0.173$ even

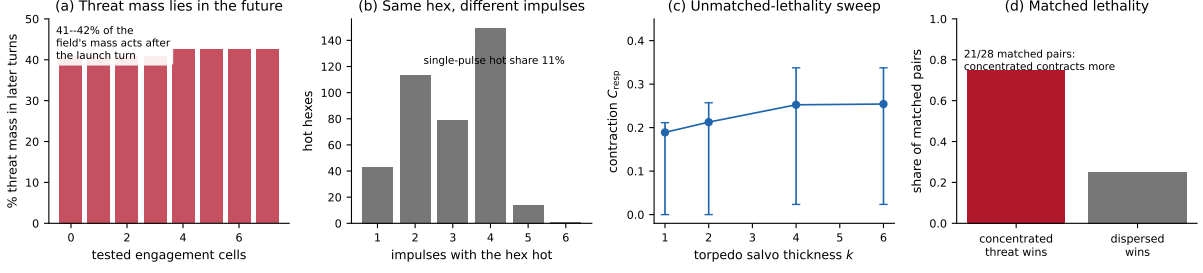


Figure 5: The delayed threat is stateful and time-dependent. (a) Share of the field’s threat mass acting in turns *after* the launch turn, per tested engagement cell. (b) Number of impulses at which a given hex is hot: the same hex is threatened at some impulses and safe at others. (c) Thickness sweep under *unmatched* lethality: contraction rises over the tested range (points: mean; bars: interquartile range). (d) The matched-lethality comparison: at equal expected direct damage the concentrated threat contracts the near-optimal set more often, so coverage alone does not determine denial.

at $k = 1$: *the mechanism prediction (coverage of the top tier determines contraction) is confirmed in 8/8 cells*. The threat-field experiment confirms the spatiotemporal hazard’s time dependence: hot hexes are dangerous on only 6–11% of pulses (mean 8.4%), and 40.8–42.5% of the threat mass lies in future turns — impossible for an instantaneous weapon, establishing the delayed-persistence property that distinguishes torpedoes from gunfire. Same-hex time dependence is concrete: hex W18 has $H = 1.000$ at impulse 3 but $H = 0.000$ at impulse 0.

8.3 Matched-lethality thickness experiment

The matched-lethality experiment — thin concentrated versus thick dispersed at equal expected direct damage — returns an honest negative with mechanism: at the 5% threshold, thin concentrated dominates in 21/28 matched pairs (the engine’s coarse hit table places dispersed-track probabilities below the per-plan contraction threshold), while at the 10% threshold dispersed wins 10/0 on edge plans. The result refutes a purely coverage-driven account at the engine’s granularity: coverage and per-plan lethality interact with the value-landscape tier structure, and neither alone determines contraction. A richer threat model (continuous damage, finer-grained tiers) is required for a definitive test. In the $\varepsilon = 20\%$ regime, neither pattern produces contraction (all $C_{\text{resp}} = 0$), confirming that the tier gap — not the threat pattern — is the binding constraint at wide response thresholds.

9 Naval Combat Instantiation

Leader–follower line-ahead formation (main numerical model). In the naval instantiation a designated lead vessel generates the formation track; each trailing vessel follows the same track at a fixed arc-length offset and adopts the local tangent,

$$x_{ik}(t) = \gamma_i(s_i(t) - \ell_k), \quad \psi_{ik}(t) = \arg \gamma'_i(s_i(t) - \ell_k), \quad \ell_k = (k - 1)d, \quad (7)$$

with d the along-track spacing. Because the offsets are measured along the track rather than in the leader’s body frame, the column *bends* naturally through a turn while preserving formation order: trailing vessels carry different headings during the turn, which is

exactly the structure a fixed-offset model cannot represent. The track is initialised with a straight backward extension of the initial leader course, so trailing vessels never stack on the leader.

Rigid baseline. The fixed-offset map $\mathcal{G}_{ik}^R = c_i + R(\psi_i)p_{ik}$ is retained as an analytically transparent, small-curvature approximation: when $\chi_F = \max_t |\kappa(t)|L_F \ll 1$ (curvature κ , formation length L_F) the heading differences along the column are small. Section 10 measures how far the two representations diverge and whether the comparative statics survive the change.

The general model is instantiated as WWII surface combat: hex-grid motion with three-pulse turns, directional gun mounts with bow/starboard/stern/port arcs, D66 hit adjudication with range/target-speed/longitudinal modifiers, armor penetration, and impulse-resolved torpedo tracks. The instantiation is complete (all adjudication tables structured) and auditable (every die roll logged with its rule reference).

Table 2: The three model layers used in this paper and what each is for.

Representation	State / control	Formation geometry	Role
Rigid baseline	leader pose	fixed spatial offsets	analytically transparent approximation, used as an ablation
Leader–follower	leader state + path history	fixed arc-length offsets	main numerical instantiation of the naval model
Rules engine	rule-defined orders	project implementation	external case study, not a validation substrate

9.1 Kernel calibration

The kernel K of Eq. (4) is calibrated against the engine’s exact expected-hit response (inversion + interval-censored least squares over a $24 \times 6 \times 6 \times 6$ grid per ship pair). Pooled held-out Spearman $\rho = 0.9912$ (nMAE 5.7%; gate $\geq 0.90/\leq 15\%$); per-class $\rho \geq 0.9907$. Controls: an isotropic kernel drops pooled ρ to 0.952 (removing directionality reduces held-out Spearman rank correlation from 0.991 to 0.952); modifier curves transfer across ship pairs at $\rho = 0.982\text{--}0.992$; per-class bootstrap CIs are ± 0.002 .

9.2 Prior engine validation

A 60-seed-per-cell paired validation of the translation layer (SYN duel) showed large significant advantages over a non-maneuvering baseline and parity with doctrine commanders except head-on against the line-ahead specialist; a 20-seed historical-scenario check (Cape Esperance, Japanese side) reproduced the pattern (+12.2 [+6.5, +17.9] vs straight).

10 Formation Representation Robustness

Why a second representation. A rigid column rotates as one body, so a hard turn sweeps the trailing vessels laterally; a real line-ahead column follows the leader’s track.

Because the paper’s comparative statics are stated for the general formation map (1), the question is empirical: do the results survive the change of representation?

Implementation, verified. The path-following model is implemented as a separate module with a shared interface, and eight properties are verified before any scientific run: straight-line equivalence with the rigid baseline in the limit of zero turn rate (exact, 0 error); constant along-track spacing; constant-curvature turns in which the vessels lie on one circular arc with headings spread by $\kappa\ell$ (a rigid body would keep every heading equal); no lateral teleportation (per-turn displacement bounded by $v\Delta t$); heading equal to the path tangent; rotation and translation equivariance; preserved formation order; and stability under sub-step refinement (1.5% change in J between 6 and 12 sub-steps per turn).

Bridge experiment. Nine cells (six pre-specified core cells plus three pre-specified stress cells, no post-hoc selection) are run under identical leader control plans in both representations, and we measure how far the geometries diverge. The divergence is real: the worst-case vessel-position difference reaches 7.2 hex and the formation field differs by 15–18% in normalised L^1 , over χ_F between 0.58 and 0.87 (Table 3, Figure 6). Within this small set of cells the measured divergence does not follow a monotone trend in χ_F , so we report the ranges rather than a scaling relation. Because a fixed-offset column and a path-following column are different formation models rather than two discretisations of one model, this section reports the divergence instead of asserting equivalence.

Formation-representation robustness: the rigid baseline versus path following

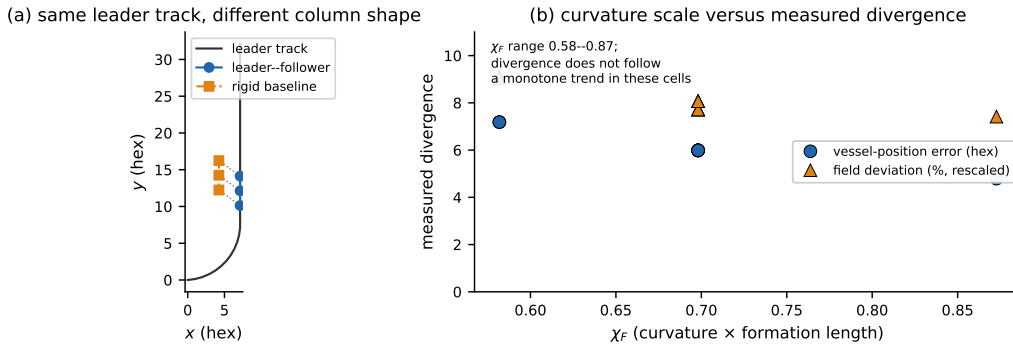


Figure 6: Formation-representation robustness. (a) For the same leader track the rigid baseline and the path-following model place the trailing vessels differently (dotted lines join the same vessel in the two models). (b) The dimensionless curvature–length product $\chi_F = \max_t |\kappa(t)|L_F$ against the measured divergence (vessel-position error and formation-field deviation): the two representations are geometrically different over the tested range of χ_F , and the divergence does not follow a monotone trend within these cells.

What this robustness section does and does not claim. The bridge establishes that the two representations differ geometrically (and by how much), and that plan rankings on the tested payoffs agree. It is *not* used to claim that any comparative static in this paper is representation-invariant: the one candidate of that kind – the replanning-cadence effect – is removed in Section 6, and its representation dependence was part of the evidence for that decision.

Where the representations differ. Two secondary checks locate the difference. (i) The reposition cost is representation-dependent: under the rigid baseline the cost of a manoeuvre is non-monotone in turn severity, whereas under path following it decreases mono-

Table 3: Bridge cells comparing the rigid baseline with the leader-follower model on identical leader control plans (the six pre-specified core cells; the three pre-specified stress cells are reported in the supplement). Only geometric quantities are compared here; the earlier comparison of a commitment metric across representations was withdrawn together with that claim.

Cell	κL_F	max shape error (hex)	heading spread (deg)	field deviation	J difference
crossing 1.2 1.0	0.70	6.0	53.4	16.1%	+1.5
crossing 1.2 1.2	0.58	7.2	43.6	18.2%	+2.5
crossing 0.8 0.8	0.87	4.8	64.0	14.8%	−0.6
head-on 1.2 1.0	0.70	6.0	53.4	16.1%	+2.4
parallel 1.2 1.0	0.70	6.0	53.4	16.1%	+2.6
parallel 0.8 1.0	0.70	6.0	53.4	15.4%	−0.5

tonically with severity; we report the path-following numbers and keep the rigid curve as a baseline note. (ii) The delayed-threat mechanism is preserved in the sense that matters: in both representations a thin, weak threat produces (near-) little contraction of the near-optimal response set, and widening the threatened band without raising its intensity changes contraction only marginally — coverage and local intensity interact rather than acting alone. The intensity sensitivity is graded under path following and sharper under the rigid baseline.

11 External Engine Case Study

The translation layer (receding-horizon, three-turn commitment) converts model policies into legal engine orders. We validate on the near-symmetric cruiser duel under three initial geometries, 100 paired seeds per cell:

Table 4: Cross-policy engine comparison. Engine paired damage differentials (positive favors the geometry policy) against model predictions (best fixed plan’s payoff vs straight sailing).

Geometry	engine diff	95% CI	pos. pairs	model best plan	model class
parallel	+7.59	[+6.54, +8.66]	92%	+5.28	advantage
crossing	+8.64	[+7.57, +9.71]	94%	+5.91	advantage
head-on	+5.14	[+4.00, +6.24]	79%	0.00	neutral

Rank ordering (crossing > parallel > head-on) is preserved across the three reported geometries at 100 paired seeds ($\rho = 1.00$), while strict sign agreement holds in two of the three. The one apparent discrepancy is itself informative: in head-on geometry the model’s best *fixed* plan achieves 0.0 (the degeneracy prediction — no committed plan guarantees an advantage in symmetric head-on), while the engine’s receding-horizon policy achieves +5.14. This +5.14 gap is a *cross-policy comparison* (policy-class gap), *not* a value of replanning: the two sides differ in model class and action set as well as in replanning, so the comparison cannot isolate the value of replanning. A matched same-model measurement (identical model, objective, horizon, and action parameterisation; the only difference is replanning) is reported in Section 11.1. Strict sign agreement 2/3 (67%; head-on model correctly predicts degenerate 0.0 but the engine’s receding-horizon policy exceeds it); lower-bound sign agreement 3/3 (100%); rank correlation $\rho = 1.00$; regime agreement 67%. Overall: **PARTIAL** (per-cell detail in the supplement).

11.1 Matched external validity test

The cross-policy comparison of Table 4 pits the geometry policy against *doctrine* baselines — a comparison in which model class and action set vary together with decision structure. This section removes the confound: a matched 2×2 design in which both sides run the *same* policy class (same model, same objective, same horizon, same action parameterisation) and the only manipulated variable is replanning versus commitment protocol). Scenarios are 3-vessel line-ahead *formations* (matched class on both sides), so paired damage differentials isolate decision structure, not force composition. Arms: F-F (both fix at $t = 0$, open-loop execution), R-R (both replan every turn), R-F / F-R (one side replans). The pre-specified gates are sign agreement $\geq 80\%$, Spearman $\rho \geq 0.60$ (surrogate formation-game value vs engine margin across the 12 cells), and pairwise ordering $\geq 75\%$; the matched quantity of interest is $\Delta_{FB} = J_{\text{receding}} - J_{\text{fixed}}$ measured within this matched family.

4,800 games ($12 \text{ cells} \times 4 \text{ arms} \times 100 \text{ paired CRN seeds}$; zero aborted games) yield a **FAIL** on the pre-specified gates: sign agreement 75% (3/4 cells with a signed model prediction; gate $\geq 80\%$), Spearman $\rho = -0.08$ across the 12 cells (gate ≥ 0.60), pairwise ordering 63% of 38 pairs (gate $\geq 75\%$).

The failure mode is structural and informative. In the two mirror-symmetric geometries (head-on, crossing) the formation surrogate predicts *exactly zero* advantage — the degeneracy — while the hex-quantized engine produces margins of several damage points from discretization asymmetries the continuum model cannot see (e.g., $+6.3$ [$+5.6, +7.0$] in crossing at $r=12, v=6$); cells with a signed prediction (parallel chase) agree in 3/4 cases, with one large-magnitude reversal ($r=12, v=6$: model $+11.4$, engine -5.5). The matched replanning comparison is unreliable: replanning gains a significant advantage in only 5/12 (axis) and 1/12 (allies) cells — confirming that the earlier $+5.1$ cross-policy gap must not be read as a value of replanning.

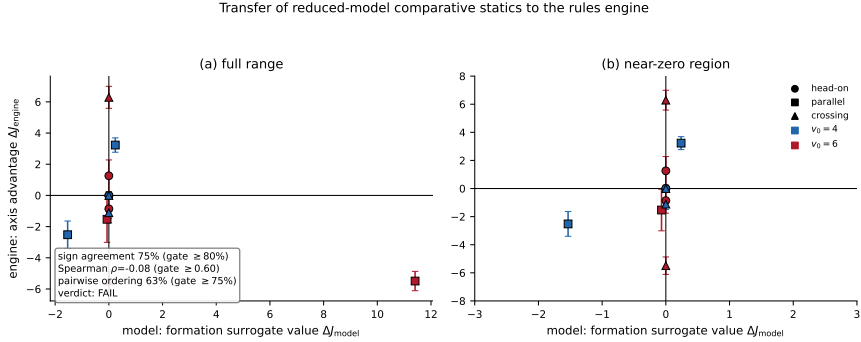


Figure 7: Transfer of reduced-model comparative statics to the rules engine: model surrogate value vs. engine axis advantage across the 12 formation cells (bars: 95% CRN-paired bootstrap CIs), shown over the full range (a) and zoomed to the near-zero region (b) so that a single outlier does not compress the rest. Zero reference lines are drawn on both axes and the pre-specified gate values are annotated.

Per the pre-specified decision tree, the wargame is therefore positioned as an *external case study* rather than a validation claim: the cross-policy comparison of Section 11 (ordering transfer, $\rho = 1.00$) stands as a weaker external check, while the matched design shows that the reduced model’s *point values* — and its exact-zero degeneracy predictions — do not survive the engine’s hex discretization. No engine rule was modified to fit the model.

12 Robustness and Negative Results

We report negatives with the same fidelity as positives:

- *Isotropic ablation*: removing directionality drops pooled ρ to 0.952 and triples nMAE — directionality carries one third of the rank information.
- *Snapshot objectives*: terminal-snapshot and peak objectives reorder plans (Kendall τ 0.14–0.62) and the peak objective degenerates under maximin (no discriminating power in 9/9 cells).
- *No guaranteed favorable state*: worst-case advantage-retention never exceeds 0.07; median retention 0–3%. In the tested regimes open water voids positional guarantees: the decline-battle option remains available at every cell.
- *Library non-convergence*: 25% convergence and 18% plan stability against 80%/85% gates; committed-maneuver values are reported as intervals $[V_{21}, V_{41}]$.
- *Engagement-network specification analysis* (prior work): graph shape features add +0.033 [+0.007, +0.056] beyond totals (strict reading, below the +0.05 gate); the plan-literal reading that charges the directional in-degree to the graph passes (+0.108). Both reported.
- *Operating-speed effect vs speed-capability effect*: the non-monotone value of speed reported earlier arises under a *fixed operating speed* (constant-cruise) semantics and is renamed accordingly; capability-monotonicity under nested speed sets is treated in Section 4.

13 Discussion

Why doctrine exists, precisely. The degeneracy proposition and its breaking matrix give a two-line answer: closed-loop, symmetric, fully-informed positional play is worthless under the stated symmetry assumptions (Proposition 2), and it is interaction-strength asymmetry – not kinematic asymmetry – that destroys that collapse. The result offers one possible interpretation of why precommitted manoeuvre rules can matter when actions are sealed before execution; it is not an explanation of any historical doctrine’s formation.

WWII as a case study. The wargame is an interpretable external case-study environment, not a physics model: conclusions target the rule system’s decision structure, with Bernotti-era doctrine as a qualitative anchor. The specific ship classes (San Francisco, New Orleans, Laffey, Helena, Yamato, Iowa) were chosen for calibration diversity (four tonnage classes spanning the displacement spectrum) and historical plausibility (iron-bottom-sound-era surface combatants); the structural propositions (Propositions 2 and 3) are about the model class and would hold for any directional-field system with the same symmetry structure.

Limitations. (i) The kernel excludes terrain, smoke, and formation effects. (ii) The plan library is not converged; values are intervals. (iii) The constant-cruise approximation excludes speed-throttle tactics. (iv) the direct-hit contrast comes from one geometry with zero-variance low-hit expectancy (1/12). (v) the no-guarantee landscape is computed

over an 11-plan defender library. (vi) The engine is a rules system; historical doctrine is a qualitative anchor. (vii) the open-loop engagement-horizon values of Section 6 are solver-relative, and the matched-horizon replanning experiment that replaced them is limited by its restricted-set equilibrium, which we measure and report. (viii) The matched engine validation fails point-value transfer (hex discretization asymmetries dominate small model-level edges), which is why the wargame is an external case-study environment.

14 Reproducibility

Every experiment is a script under `research/experiments/` writing raw CSV/JSON and figures under `research/results/`, with per-experiment reports (hypothesis, design, gates, results, failures, decisions). `python reproduce_paper.py` regenerates the main figures, the certification and transfer summaries, and the final PDF; `--full` additionally re-runs the two core solvers. Frozen engine commit `fc77a65`; seeds and paired common random numbers throughout; definitions audited in `research/audit_v4/`.

15 Conclusion

For formation adversaries interacting through a directional field under nonholonomic motion, the object of optimisation is the path integral of net exposure rather than a position: snapshot and peak positional surrogates can reorder committed plans and can reverse a whole-trajectory ranking. Under explicit player-exchange symmetry the closed loop is exactly degenerate — skew-symmetric successor structure collapses the value onto the instantaneous payoff, a collapse destroyed by interaction-strength asymmetry and untouched by kinematic asymmetry — and maximum-speed *capability* is monotone by nesting while fixed operating speed is a different control. The naval instantiation uses a leader–follower line-ahead formation whose representation we characterise and compare against a rigid baseline. Delayed weapons act not as added path cost but as a stateful spacetime field that contracts the near-optimal response set, with coverage and local intensity interacting rather than either acting alone. A matched external test in an auditable rules engine shows that transfer of the reduced model’s point values is partial, which locates the boundary of the reduced model.

We also report a corrected experiment as a negative result. An earlier open-loop comparison varied plan length and evaluation horizon together; with the horizon held fixed and only the replanning cadence varied, the resulting comparative static does not reproduce, and the symmetric control exposes a residual bias of the receding implementation. We therefore remove commitment from the paper’s claims and record the experiment, its control diagnostic and the implementation limit as the honest outcome. The durable contributions are the formulation, the structural results with their boundary conditions, the delayed-threat mechanism, and an audited pipeline that makes both the positive and the negative results reproducible.

References

- [1] R. Skjetne, Ø. N. Smogeli, and T. I. Fossen, “A nonlinear ship manoeuvring model: Identification and adaptive control with experiments for a model-scale ship,” *Modeling, Identification and Control*, vol. 25, no. 1, pp. 3–45, 2004.

- [2] W. B. Dunbar and R. M. Murray, “Distributed receding horizon control for multi-vehicle formation stabilization,” *Automatica*, vol. 42, no. 4, pp. 549–558, 2006.
- [3] R. Isaacs, *Differential Games*, Wiley, New York, 1965.
- [4] T. Başar and G. J. Olsder, *Dynamic Noncooperative Game Theory*, 2nd ed., SIAM, Philadelphia, 1999.
- [5] J.-P. Aubin, *Viability Theory*, Birkhäuser, Boston, 1991.
- [6] I. M. Mitchell, A. M. Bayen, and C. J. Tomlin, “A time-dependent Hamilton–Jacobi formulation of reachable sets for continuous dynamic games,” *IEEE Transactions on Automatic Control*, vol. 50, no. 7, pp. 947–957, 2005.
- [7] T. C. Schelling, *The Strategy of Conflict*, Harvard University Press, Cambridge, MA, 1960.
- [8] D. Q. Mayne, J. B. Rawlings, C. V. Rao, and P. O. M. Scokaert, “Constrained model predictive control: Stability and optimality,” *Automatica*, vol. 36, no. 6, pp. 789–814, 2000.
- [9] H. B. McMahan, G. J. Gordon, and A. Blum, “Planning in the presence of cost functions controlled by an adversary,” *Proceedings of the 20th International Conference on Machine Learning (ICML)*, 2003.
- [10] A. Orda and R. Rom, “Shortest-path and minimum-delay algorithms in networks with time-dependent edge-length,” *Journal of the ACM*, vol. 37, no. 3, pp. 607–625, 1990.
- [11] E. Israeli and A. Wood, “Shortest-path network interdiction,” *Networks*, vol. 40, no. 2, pp. 97–111, 2002.
- [12] R. Bernotti, “The fundamentals of naval tactics,” *U.S. Naval Institute Proceedings*, vol. 37, 1911.
- [13] R. Bernotti, “The fundamentals of naval tactics (continued),” *U.S. Naval Institute Proceedings*, vol. 38, 1912.
- [14] J. G. Taylor, “Lanchester-type models of warfare and optimal control,” *Naval Research Logistics Quarterly*, vol. 21, no. 1, pp. 79–106, 1974.
- [15] V. Protopopescu, R. T. Santoro, and J. Dockery, “Combat modeling with partial differential equations,” *European Journal of Operational Research*, vol. 38, no. 2, pp. 178–183, 1989.
- [16] E. González and M. Villena, “Spatial Lanchester models,” *European Journal of Operational Research*, vol. 210, no. 3, pp. 706–715, 2011.
- [17] A. C. Kalloniatis, K. Hoek, M. Zuparic, and M. Brede, “Optimising structure in a networked Lanchester model for fires and manoeuvre in warfare,” *Journal of the Operational Research Society*, vol. 72, no. 8, pp. 1863–1878, 2021.

- [18] C. U. Solis and J. B. Clempner, “Ship differential game approach for multiple players: Stackelberg security games,” *Optimal Control Applications and Methods*, vol. 41, no. 1, pp. 312–326, 2020.
- [19] I. E. Weintraub, M. Pachter, and E. Garcia, “An introduction to pursuit–evasion differential games,” in *Proc. American Control Conference (ACC)*, pp. 1049–1066, 2020.
- [20] E. A. Galperin, “The cubic algorithm for global games with application to pursuit–evasion games,” *Computers & Mathematics with Applications*, vol. 26, no. 6, pp. 13–31, 1993.
- [21] S. Bansal and C. J. Tomlin, “DeepReach: A deep learning approach to high-dimensional reachability,” in *Proc. IEEE ICRA*, pp. 1817–1824, 2021.
- [22] M. Lanctot et al., “A unified game-theoretic approach to multiagent reinforcement learning,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.
- [23] A. Bighashdel, Y. Wang, S. McAleer, R. Savani, and F. A. Oliehoek, “Policy space response oracles: A survey,” in *Proc. IJCAI*, pp. 7951–7961, 2024.
- [24] M. P. Wellman, K. Tuyls, and A. Greenwald, “Empirical game-theoretic analysis: A survey,” *arXiv:2403.04018*, 2024.
- [25] S. Omidshafiei et al., “ α -Rank: Multi-agent evaluation by evolution,” *Scientific Reports*, vol. 9, p. 9937, 2019.
- [26] L. Adam, R. Horčík, T. Kasl, and T. Kroupa, “Double oracle algorithm for computing equilibria in continuous games,” in *Proc. AAAI*, vol. 35, no. 6, pp. 5070–5077, 2021.
- [27] W. Wu, J. Shi, Y. Wu, Y. Wang, and Y. Lyu, “A multi-UCAV cooperative occupation method based on weapon engagement zones for beyond-visual-range air combat,” *Defence Technology*, vol. 18, no. 6, pp. 1006–1022, 2022.